GetStandardDeviation [SimTalk]

Returns the standard deviation of the values that the *Display* designated by <Path> recorded.

Type

Read-only attribute

Syntax

```
<Path>.GetStandardDeviation → real
```

Return Value

The return value has the data type *real*.

Example

```
print MyDisplay.GetStandardDeviation
```

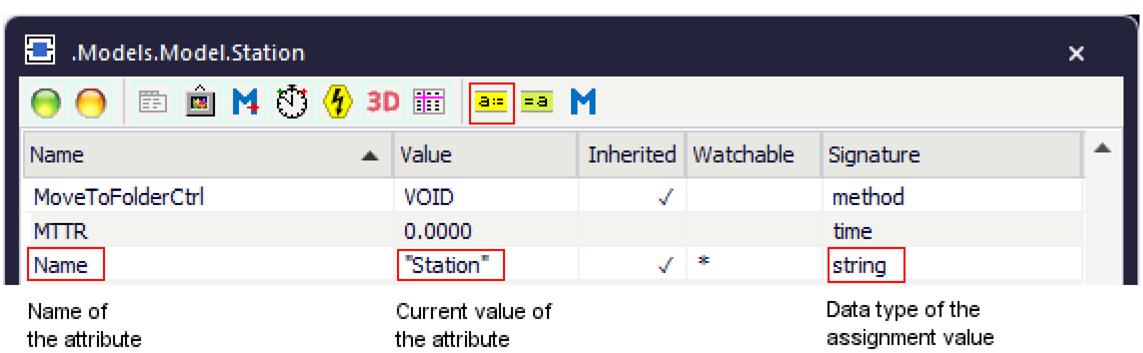
Attributes of the Display

_Attributes of the Display

The *Display* provides:

- The *attributes* listed in the table of contents to the left.
- The **Attributes of All Objects**.

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window **Show Attributes and Methods**. The figure below illustrates the information using the example of the object *Station*.



- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected **Class [general description]**.
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected **Instance [general description]**.

You can set the value of an attribute and you can get its value, either with the check boxes, the text boxes and drop-down lists in the dialog windows or by assigning values to the respective attributes.

- To **set the value** of an attribute, you might, for example, type:

```
Display1.Sampler := false
Display1.Path := "MyConveyor.NumMU"
Display2.Sampler := true
Display2.SmpInterval := 300
Display2.Path := "root.MyConveyor.utilization"
```

- To **get the value** of an attribute, you might, for example, type:

```
print Display1.Path
posit := Station.Cont.XPos
```

Active [SimTalk] - Display

Activates the object designated by <Path> (`true`) or deactivates it (`false`).

Type

Attribute

Syntax

```
<Path>.Active:boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Example

```
MyDisplay.active := true
```

See also

Active [check box] - Display

Color [SimTalk] - Display

Sets the **Color** for displaying the values which the object *Display* designated by <Path> visualizes in the *Frame*.

Remarks

Set the RGB values of the color with the method *makeRGBValue*.

Type

Attribute

Syntax

```
<Path>.Color:integer
```

Assignment Value

You can assign a value of data type *integer*.

Example

```
MyDisplay.Color := makeRGBValue(255,0,0)
```

SimTalk

makeRGBValue [SimTalk]

See also

Color [Display]

Comment [SimTalk] - Display

Sets the **Comment** that the *Display* designated by <Path> shows underneath its icon in the *Frame*.

Type

Attribute

Syntax

```
<Path>.Comment:string
```

Assignment Value

You can assign a value of data type *string*.

Example

```
MyDisplay.Comment := "My comment"
```

See also

Comment [text box] - Display

DecimalPlaces [SimTalk] - Display

Sets the number of **Decimal Places** which the *Display* designated by <Path> shows.

Remarks

The *Display* can show up to 15 decimal places. For the default setting -1 shows all existing places. Type in a positive value to show this number of places.

> **Note**
>
> This setting only applies to the **Type > Text**.

Type

Attribute

Syntax

```
<Path>.DecimalPlaces:integer
```

Assignment Value

You can assign a value of data type *integer*.

Example

```
MyDisplay.DecimalPlaces := 10
```

See also

Decimal Places [text box] - Display

Type [drop-down list] - Display

DisplayType [SimTalk]

Sets if the *Display* designated by <Path> shows the input value as **Text**, as a **Bar**, or as a **Pie Segment**.

Remarks

Bar and **Pie** representation map values in a specified range to bars of varying height or pies with a varying degree of completion.

You can set the range with the attributes *MinVal* and *MaxVal*.

Type

Attribute

Syntax

```
<Path>.DisplayType:string
```

Assignment Value

You can assign a value of data type *string*.

You can specify `"Text"`, `"Bar"`, or `"Pie"`.

Example

```
MyDisplay.DisplayMode := "Text"
```

SimTalk

MinVal [SimTalk]

MaxVal [SimTalk]

See also

Type [drop-down list] - Display

Font [SimTalk] - Display

Sets the **Font Size** of the text and of the **Comment**, which the *Display* designated by <Path> displays in **Text** mode in the *Frame*.

Type

Attribute

Syntax

```
<Path>.Font:integer
```

Assignment Value

You can assign a value of data type *integer*.

You can specify 1 for **Small**, 2 for **Medium**, 3 for **Large**, 4 for **Extra Large**.

Example

```
MyDisplay.Font := 2
```

See also

Font Size [drop-down list] - Display

Comment [text box] - Display

MaxVal [SimTalk]

Sets the **Upper Bound** of the displayed range in **Bar** and **Pie** modes of the *Display* designated by <Path>.

Remarks

The *Display* shows values beyond or equal to *MaxVal* as a full bar/pie.

Type

Attribute

Syntax

```
<Path>.MaxVal:real
```

Assignment Value

You can assign a value of data type *real*.

Example

```
MyDisplay.MinVal := 10.0
MyDisplay.MaxVal := 35.713
```

See also

Type [drop-down list] - Display

MinVal [SimTalk]

Sets the **Lower Bound** of the displayed range in **Bar** and **Pie** modes.

Remarks

The *Display* designated by <Path> shows values below or equal to *MinVal* as an empty bar/pie.

Type

Attribute

Syntax

```
<Path>.MinVal:real
```

Assignment Value

You can assign a value of data type *real*.

Example

```
MyDisplay.MinVal := 5.35
```

See also

Type [drop-down list] - Display

Path [SimTalk] - Display

Sets the **Path** to the value which the *Display* designated by <Path> shows.

Type

Attribute

Syntax

```
<Path>.Path:string
```

Assignment Value

You can assign a value of data type *string*.

Example

```
Display1.Sampler := false
Display1.Path := "MyConveyor.NumMU"
Display2.Sampler := true
Display2.SmpInterval := 300
Display2.Path := "root.MyConveyor.utilization"
```

See also

Path [text box] - Display

Sampler [SimTalk]

Activates **Sample** mode of the *Display* designated by <Path> (`true`) or deactivates it (`false`).

Type

Attribute

Syntax

```
<Path>.Sampler:boolean
```

Assignment Value

You can assign a value of data type *boolean*.

- `true` designates **Sample** mode. It updates the display periodically, independent of changes of the input value.
- `false` designates **Watch** mode. It updates its display whenever the input value changes.

Example

```
if MeinDisplay.Sampler = false
   MyDisplay.Sampler := true // activates sample mode
   MyDisplay.SmpInterval := 360 // 10 minutes
end
```

See also

Mode [drop-down list] - Display

SmpPeriod [SimTalk] - Display

Sets the intervals after which the *Display* designated by <Path> updates the display. Its value must be greater than zero.

Remarks

SmpPeriod applies to **Sample** mode.

Type

Attribute

Syntax

```
<Path>.SmpPeriod:time
```

Assignment Value

You can assign a value of data type *time*.

Example

```
MyDisplay.SmpPeriod := str_to_time("3:00.0")
MyDisplay.SmpPeriod := 2 * MyDisplay.SmpPeriod
// double interval, half the sampling rate
```

See also

Interval [text box] - Display

Transparent [SimTalk] - Display

Makes the background of the object designated by <Path> **Transparent** in the *Frame* (`true`) or not transparent (`false`).

Type

Attribute

Syntax

```
<Path>.Transparent:boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Specify `true` to show the space around the text in the color of the background image of the *Frame*.

Specify `false` to show the space around the text in white.

Example

```
MyDisplay.Transparent := false
```

Value [SimTalk] - Display

Sets or gets the **Value**, which the *Display* designated by <Path> shows.

Type

Only use the attribute *Value* for **Watch Mode**.

In **Sample Mode** *Plant Simulation* only updates the value if **MUs and States**  is activated and the *Frame*, in which the *Display* is inserted, is open.

When you get the value and the *Display* is in **Sample Mode**, *Plant Simulation* updates the value beforehand.

Type

Attribute

Syntax

```
<Path>.Value:real
```

Assignment Value

You can assign a value of data type *real*.

Return Value

The return value has the data type of the observed value.

Example

```
MyDisplay.Value := 5
print MyDisplay.Value
```

See also

Mode [drop-down list] - Display

Value [Display]

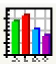
MUs and States [Home ribbon]

Chart

Chart [object]

Use the object *Chart* for presenting current data and results of the simulation run.

Image



Description

Define data to be plotted:

- By using a **Table** containing the data, such as simulation results.
- By defining **Input Channels** that record values of attributes of interest of the objects.

A *Chart*, which you insert into a *Frame*, provides the **Statistics Wizard**. Select the command **Statistics Wizard** on the context menu of the *Frame* to open it.